SE3002 Software Quality Engineering

Assignment 2: Final Testing Report

Student Course Registration System (registration.py)

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GitHub repository: https://github.com/moodi-mrt/SQE-Assignment-2

1. Black-box test design

All black-box cases were written from REQUIREMENTS.md before reading the branches in registration.py. Expected values were never changed to match the code. A baseline student (S001, 12 credits, no dues, passed CS101) and a baseline course (CS201, 3 credits, 10/30 seats, needs CS101) are used, and each case only changes the values it is about. Full sheets are in docs/blackbox_test_design.xlsx and docs/decision_table.xlsx.

1.1 Equivalence class partitioning (19 cases)

Attribute	Valid classes	Invalid classes	Test IDs
Credit hours (FR-01/02)	current 0-18 and course 1-4 with total <= 18; total > 18 (returns False)	current < 0, current > 18, course < 1, course > 4, both invalid (order check)	ECP-01..06, ECP-17
Outstanding dues (FR-04/15)	dues = 0	dues > 0 (DUES_BLOCK), dues < 0 (ValueError), negative dues + bad capacity (order)	ECP-07..09, ECP-19
Prerequisites (FR-03/06)	none required, all completed	some completed, none completed	ECP-10..13, ECP-18
Registration status (FR-05)	not registered	already registered	ECP-16
Availability (FR-08)	enrolled < capacity	enrolled == capacity (WAITLISTED)	ECP-14, ECP-15

1.2 Boundary value analysis (23 cases)

Variable	Values tested	Test IDs
current credits (0-18)	-1, 0, 18, 19	BVA-03, 01, 02, 04
course credits (1-4)	0, 1, 4, 5	BVA-07, 05, 06, 08
combined total (max 18)	17, 18, 19 (helper) and 18 through registration_decision	BVA-09, 10, 11, 22
enrolled vs capacity 30	-1, 0, 29, 30, 31	BVA-16, 15, 12, 13, 14
capacity (>= 1)	0, 1 (empty), 1 (full)	BVA-18, 17, 23
dues near zero	-1, 0, 1	BVA-21, 19, 20

1.3 Decision table for registration_decision()

Five conditions give 32 combinations, but because of the FR-09 priority order (dues, already registered, prerequisites, credit limit, availability) they reduce to six rules. Rules R1-R4 each also got a priority test where every lower-priority condition is true as well, to check that only the first applicable outcome is returned.

	R1	R2	R3	R4	R5	R6
C1 dues > 0	T	F	F	F	F	F
C2 already registered	-	T	F	F	F	F

	R1	R2	R3	R4	R5	R6
C3 prerequisite missing	-	-	T	F	F	F
C4 total credits > 18	-	-	-	T	F	F
C5 course full	-	-	-	-	T	F
Outcome	DUES_BLOCK	ALREADY_ REGISTERED	PREREQUISITES_ MISSING	CREDIT_ LIMIT	WAITLISTED	REGISTERED
Basic test	DT-01	DT-02	DT-03	DT-04	DT-05	DT-06
Priority test	DT-07	DT-08	DT-09	DT-10	-	-

1.4 State change tests (13 cases)

ST-01..ST-13 check FR-10 to FR-14: a successful registration updates registered_courses, credits and enrolled; WAITLISTED only appends the ID once; every other outcome (and ValueError) leaves both objects equal to a deep copy taken before the call; withdrawal reverses a registration; withdrawing a course the student does not hold returns False and changes nothing.

1.5 Suite optimisation (A5)

We compared the call, the input values and the expected result of every case. Only cases where all three were the same were treated as duplicates. They are executed once, and the pytest id keeps every design ID, so traceability is not lost (e.g. `test_registration_decision[ECP-15|BVA-13|DT-05]`).

Merged group	Shared case
ECP-07 | ECP-14 | BVA-19 | DT-06	baseline student + baseline course -> REGISTERED
ECP-08 | DT-01	dues 500 -> DUES_BLOCK
ECP-15 | BVA-13 | DT-05	enrolled 30 / capacity 30 -> WAITLISTED
ECP-16 | DT-02	already registered in CS201 -> ALREADY_REGISTERED
ECP-06 | BVA-07	check_credit_limit(10, 0) -> ValueError

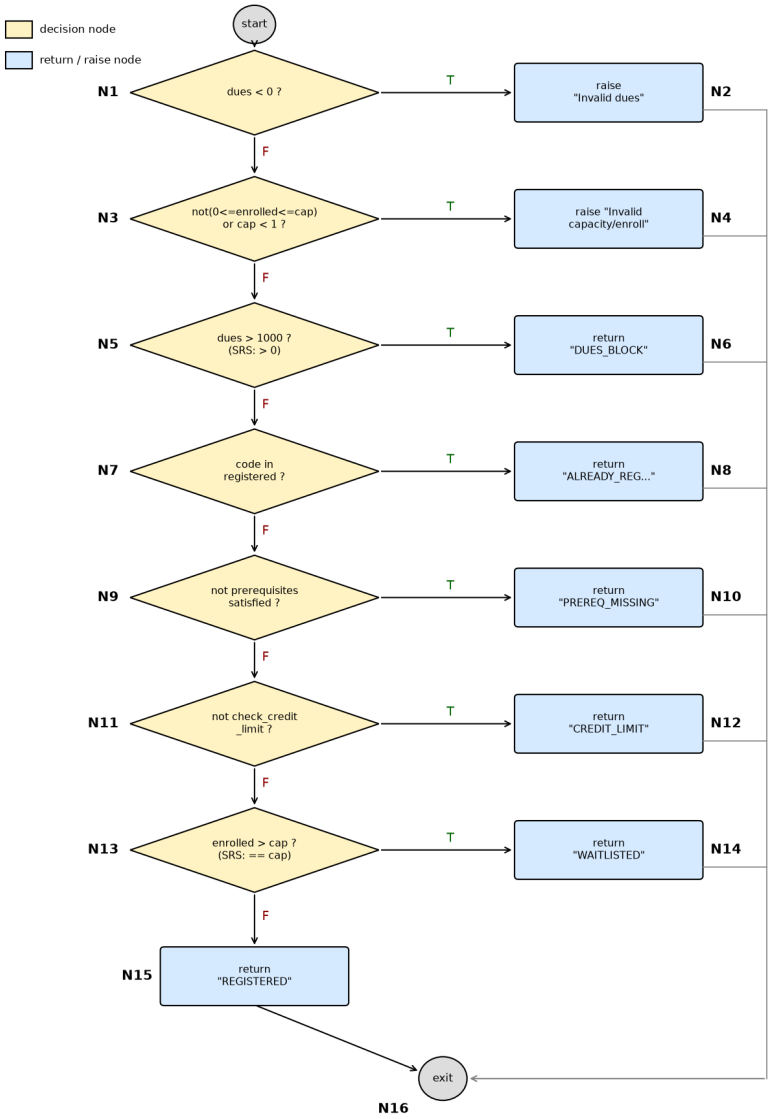
Not merged even though the output is the same: BVA-20 (dues 1) and ECP-08 (dues 500) use different inputs; BVA-11 tests check_credit_limit while DT-04 tests registration_decision; BVA-10, BVA-22 and ST-11 all use 15 + 3 credits but check the helper result, the status and the state change respectively. Result: 65 designed cases, 57 executed.

1.6 Initial black-box results (original code)

57 executed: 42 passed, 15 failed (18 design IDs, because some failing cases are merged groups). Failing: BVA-10, ECP-12, ECP-08|DT-01, BVA-20, ECP-18, BVA-22, ECP-15|BVA-13|DT-05, BVA-23, DT-07, ST-02, ST-03, ST-11, ST-04, ST-08, ST-13. These were traced to five separate defects (section 4).

2. White-box analysis

2.1 Control flow graph of registration_decision()




```
registration_decision()  -  Control Flow Graph (original code)
Nodes N = 16 (7 decision, 8 return/raise, 1 exit; entry merged into N1)
Edges E = 22  ->  V(G) = E - N + 2 = 22 - 16 + 2 = 8
Predicate nodes P = 7  ->  V(G) = P + 1 = 8   (compound N3 counted as one node)
If N3 is split into its 3 short-circuit atomic conditions: P = 9, V(G) = 10
```

2.2 Cyclomatic complexity

Nodes: 7 decision nodes (N1, N3, N5, N7, N9, N11, N13), 8 return/raise nodes (N2, N4, N6, N8, N10, N12, N14, N15) and one exit node N16, so N = 16. Each decision node has two outgoing edges (14) and each return/raise node has one edge to the exit (8), so E = 22.

V(G) = E - N + 2 = 22 - 16 + 2 = 8.

Convention for the compound condition: in the CFG, N3 (`not 0 <= enrolled <= capacity or capacity < 1`) is drawn as one predicate node. With P = 7 predicate nodes, V(G) = P + 1 = 8, which agrees with E - N + 2. If N3 is expanded into its three short-circuit atomic conditions (0 <= enrolled, enrolled <= capacity, capacity < 1) it becomes three predicate nodes, so P = 9 and V(G) = 10. The 8-path basis set below uses the collapsed graph; the extra two paths of the expanded graph are covered by the

MC/DC tests in 2.4. Exceptions raised inside check_credit_limit are not drawn because valid fixtures never reach them from here.

2.3 Independent basis paths

Path	Node sequence	Input	Expected	Test
P1	N1(T) N2	dues -1	ValueError Invalid dues	WB01
P2	N1(F) N3(T) N4	enrolled 21 / cap 20	ValueError capacity	WB02
P3	N1 N3 N5(T) N6	dues 1500	DUES_BLOCK	WB03
P4	... N5(F) N7(T) N8	already registered	ALREADY_REGISTERED	WB04
P5	... N7(F) N9(T) N10	no completed courses	PREREQUISITES_MISSING	WB05
P6	... N9(F) N11(T) N12	current 18	CREDIT_LIMIT	WB06
P7	... N11(F) N13(T) N14	enrolled 20 / cap 20	WAITLISTED	WB07
P8	... N13(F) N15	baseline	REGISTERED	WB08

P7 is **infeasible in the original code**: N13 tested enrolled > capacity, but N3 already raises for that state, so N14 could never be reached. This is the same fault black-box test ECP-15|BVA-13|DT-05 found. After the fix (enrolled >= capacity) P7 is feasible and WB07 passes. P3 uses dues of 1500 so the path is exercised regardless of the threshold; black-box tests used 1 and 500 and never reached line 51.

2.4 Condition coverage and MC/DC

Atomic conditions in registration.py: dues < 0; A = 0 <= enrolled, B = enrolled <= capacity, C = capacity < 1; dues > 0; code in registered; not required (helper); check_credit_limit range checks; enrolled >= capacity; and in register_course D = decision == 'WAITLISTED', W = id not in waitlist. Single-condition decisions get condition coverage from the basis path tests (both T and F seen). MC/DC was done by hand for the two compound decisions because pytest-cov only reports lines and branches.

Decision N3: (not (A and B)) or C. Python stops at A=False (B not evaluated) and skips C when the left side is already True. x = not evaluated.

Test	capacity	enrolled	A	B	C	Decision	Result
MC1	20	5	T	T	F	F	REGISTERED
MC2	20	-1	F	x	x	T	ValueError
MC3	20	21	T	F	x	T	ValueError
MC4	0	0	T	T	T	T	ValueError
WB10	-3	0	T	F	x	T	ValueError

Independence pairs: A by MC1/MC2, B by MC1/MC3, C by MC1/MC4. In each pair only the condition under test changes among the evaluated conditions and the decision flips. Observation: C can only affect the result when capacity = enrolled = 0; for any negative capacity B is already False (WB10), so C is almost redundant, but it is not removable because of the 0/0 case.

Decision in register_course: D and W (W not evaluated when D is False).

Test	Situation	D	W	Append?
WB11	full course, ID not on list	T	T	yes
WB12	full course, ID already on list	T	F	no
WB13	CREDIT_LIMIT outcome	F	x	no

D is shown by WB11/WB13 and W by WB11/WB12. On the original code D could never be True, so these tests failed until defect #4 was fixed.

2.5 Statement and branch coverage

Run	Statements	Branches	Combined	Missing lines	Result
Black-box only, original code	54/57 (94.7%)	23/26 (88.5%)	92.8%	51, 59, 71	42 pass / 15 fail
Black-box + white-box, original code	55/57 (96.5%)	24/26 (92.3%)	95.2%	59, 71	60 pass / 22 fail
Black-box + white-box, after fixes	58/58 (100%)	26/26 (100%)	100%	-	82 pass / 0 fail

Gaps in the first run: line 51 (return DUES_BLOCK, branch 50->51) because no black-box value of dues was above the coded threshold of 1000; line 59 (return WAITLISTED, branch 58->59) which is unreachable; line 71 (waitlist append, branch 70->71) which depends on line 59. WB03 and WB22 closed line 51. Lines 59 and 71 stay uncovered on the original code because they cannot be reached, and this is reported as a defect rather than hidden. After the fixes the statement count is 58 (one line added in withdraw_course) and everything is covered. Added white-box test IDs: WB01-WB22 (25 pytest cases including parametrised MC1-MC4 and WB18/WB19).

2.6 White-box test targets

Test	Target
WB01-WB08	basis paths P1-P8 of registration_decision
MC1-MC4, WB10	MC/DC and short-circuit of the N3 capacity/enrollment condition
WB11-WB14	MC/DC of the waitlist condition in register_course, list order
WB15-WB17	both branches of prerequisites_satisfied incl. partial overlap
WB18, WB19	True and False outcome of the final comparison in check_credit_limit
WB20, WB21	both branches of withdraw_course
WB22	register_course where neither the REGISTERED nor the WAITLISTED branch runs

3. Defects

Issue	Req	Defect	Expected / actual	Failing tests	Status
#1	FR-01	check_credit_limit uses < 18	15+3: expected True, got False	BVA-10, BVA-22, ST-11, WB18	Fixed, closed (PR #8)
#2	FR-03, FR-06	prerequisites use intersection (any one)	{CS101} vs {CS101,MATH101}: expected False, got True	ECP-12, ECP-18, WB17	Fixed
#3	FR-04	dues only block above Rs. 1000	dues 1: expected DUES_BLOCK, got REGISTERED	ECP-08|DT-01, BVA-20, DT-07, ST-04	Fixed
#4	FR-08, FR-12	full course check uses enrolled > capacity	30/30: expected WAITLISTED, got REGISTERED (enrolled 31)	ECP-15|BVA-13|DT-05, BVA-23, ST-02, ST-03, WB07, WB11, WB12, WB14	Fixed
#5	FR-13	withdraw_course does not decrement enrolled	enrolled 11 -> expected 10, stays 11	ST-08, ST-13, WB20	Fixed

Each defect was fixed in its own commit on the defect-fixes branch (commit message references the issue). No test expectations were changed. Regression run after the fixes: 82 passed, 0 failed.

4. GitHub workflow

Item	Details
Repository	https://github.com/moodi-mrt/SQE-Assignment-2
Branches	main, blackbox-testing, whitebox-testing, defect-fixes, final-report, report-update
Issues	#1-#5, one per defect (requirement, steps, expected/actual, failing test ID, status). All closed by PR #8. https://github.com/moodi-mrt/SQE-Assignment-2/issues?q=is%3Aissue
Pull requests	#6 blackbox-testing -> main: black-box suite + CI workflow (CI red, kept as evidence) #7 whitebox-testing -> main: white-box tests + CFG (CI red, still before fixes) #8 defect-fixes -> main: five fixes, closes #1-#5 (CI green) #9 final-report -> main: this report and README #10 report-update -> main: name and roll number added to this report https://github.com/moodi-mrt/SQE-Assignment-2/pulls?q=is%3Apr
Actions evidence	Failing run before fixes (15 failed, 42 passed): https://github.com/moodi-mrt/SQE-Assignment-2/actions/runs/43432390489 Failing run with white-box tests (22 failed, 60 passed): https://github.com/moodi-mrt/SQE-Assignment-2/actions/runs/43432601720 Passing run after fixes (82 passed): https://github.com/moodi-mrt/SQE-Assignment-2/actions/runs/43432791090
CI	.github/workflows/tests.yml runs on every push and PR: installs requirements, runs the black-box suite, runs all tests with branch coverage, uploads JUnit XML, coverage XML and HTML as an artifact

5. Contributions

Individual submission (Mehmood Khan, 24P-3037, GitHub: moodi-mrt): test design (ECP, BVA, decision table), black-box and white-box test implementation, CFG and MC/DC analysis, coverage measurement, defect reports, fixes, CI workflow and this report.